KLH
BACHUPALLY

Intelligent Metro Flow Optimization Platform

PROJECT BASED LEARNING — DSA

Abstract

The **Intelligent Metro Flow Optimization Platform** is a Java-based application designed to manage and - analyze passenger travel information using Data Structures and Algorithms (DSA). System allowe users to - enter passenger details such as passenger ID, name, start station, destination, and travel time through aicon- sole interface. ~The application' processes passenger data using various data structures including arrays, linked lists, queues, and hash tables. Algorithms such as bubble sort and linear search organize, and retrieve passenger information efficiently.

This project demonstrates how fundamental Data Structures, and **Algorithms (DSA)** Con be used to optimize data processing, improve search efficiency, and manage passenger travel information effectively.

SYSTEM ARCHITECTURE

Java Console (User)

Java Console Interface

Passenger (User)

Passenger Data Processsing

Passenger System(s)

- Passenger Storage (Array)
- Passenger Management (Linked List)
- Passenger Boarding System (Queue)
- Fast Passenger Lookup (Hash Table)

Output
- Passenger List
- Passenger List — Sorted Passengers by travel time
- Linear Search — Passenger Boarding Queue

CO-WISE IMPLEMENTATION

CO1 Algorithm Analysis (Searching & Sorting)

- ✔ Modules:
- ✔ Linear Search Implementation
- ✔ Bubble Sort Algorithm
- ✔ Passenger Data Sorting

Sorted Passenger List
123 gpta Mtiypnor – Time 10
123 gpta Mtiypnor – jmne 10
100 gpta Stiypnor
100 gpta Mtiypnor – jmne 10

Sorted Passenger List
123 gpta Mtiypnor – jmna Tmne 80
123 gpta Mtiypnor – jmna Time 10

Search Passenger Result
123 gpta Mtiypnor – jmna Time 10
123 gpta Mtiypnor – jmna Time 10

Screenshot to include

Search Passenger Result

CO2 Abstract Data Types (Arrays & Linked Lists)

- ✔ Modules:
- ✔ Passenger Storage using Array
- ✔ Passenger Management using Linked List
- ✔ Insert and Display Passenger Records

Passenger Cist Dipplay
123 gpta Stiypnor – Time 10
123 gpta Mtiypnor – Time 10
100 gpta Stiypnor
123 gpta Stiypnor – jmne 10

CO3 Queue Implementation

- Modules:
- ✔ Passenger Boarding Queue
- ✔ FIFO Processing of Passengers

Hash Tiablie Seout
100 gpta Stiypnor – jmna Tmne 10
123 gpnu
100 gpta Stiypnor – jmna Tmne 10

103, gayu miyapur – jmu Time 10

CO4 Hash Based Data Structures

- Modules:
- ✔ Passenger Lookup using Hash Table
- ✔ Fast Passenger Search using Hash Function

Hash Table Seout
123 gpta Stiypnor – jmna Time 10
102 gaau
103 gpta Stiypnor – jmna Time 10

103, gayu miyapur – jmu Time 10

Future Scope

- Integration with real metro transportation systems,
- Real time passenger traffic analysis
- Web-based passenger management platform
- AI-based passenger flow prodiction

Conclusion

The **Intelligent Metro Flow Optimization Platform**demonstrates the practical application of Data Structures and Algorith, thn. by implementing arrays. linked lists. Queues, Sorting algorithms and fiash tables. the system provides structured.

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